COMPUTER SYSTEMS FUNDAMENTALS

(4COSC004W)

Module Overview

Contact details

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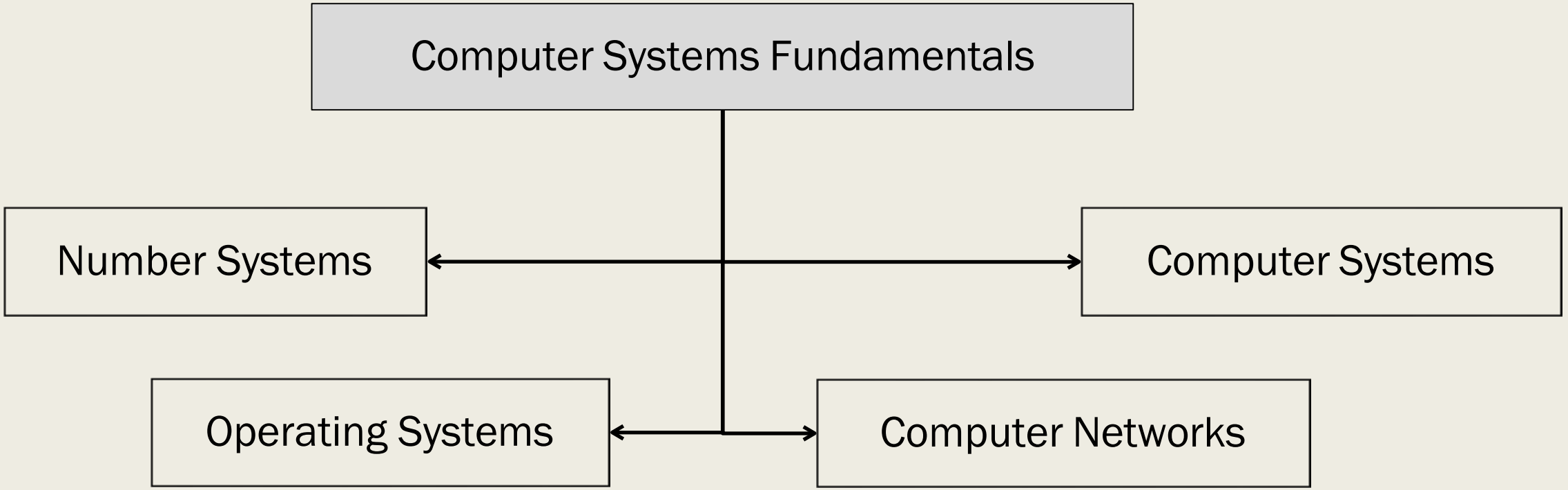
- A special thanks to Noam Weingarten

MODULE OVERVIEW

Topics in this module

- Number systems
- Logical & arithmetic operations
- Data storage
- Operating Systems
- File systems
- Assembly Language
- Process management
- Memory management
- Networking

Diagrammatic View of Module



Number Systems

- Decimal / Denary 1234567890
- Binary 10100101
- Hexadecimal BC4A0050
- Fractional
  - *Decimal* *123.456*
  - *Bicimal* *101.001 (or 5.125)*
  - *IEEE754*
- Text & Image representations
  - *ASCII & Unicode*
  - *RGB image formats*

Operating Systems & File Systems

- Types of OS
- Functions of OS
- Memory management
- Process management
- File Management
- Sector addressing
- Disk partitions & consistency
- Master Boot Record Partition Tables
- Disk Sector Zero
- Partition Sector Zero

Computer Systems & Networking

- Logical and Arithmetic Operations
- Major Hardware Components
  - *Motherboard*
  - *CPU*
  - *Memory*
  - *Hard disk*
  - *Power supply*
- Raspberry Pi
- Assembly Language
- Network Topologies
  - *Physical & Logical*
- Types of network
- Collision detection
- OSI 7-layered architecture
- IP addresses

Module Learning Outcomes

- After completing this module, you will be able to:
  - *Describe the tasks and features of each of the main components of a computer system and how these components work together.*
  - *Demonstrate a thorough and critical understanding of the methodology employed by the Operating System in managing processes, memory and the file systems including multi–threading and concurrency.*
  - *Show a thorough understanding of the von Neumann architecture and stored program paradigm.*
  - *Demonstrate a thorough and critical understanding of the various network topologies, models, protocols, interfaces, IP address configuration and network sub–netting.*
  - *Demonstrate a thorough knowledge of some simple command line commands.*

HOW IT WILL WORK

Weekly pattern:

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When is my tutorial slot?

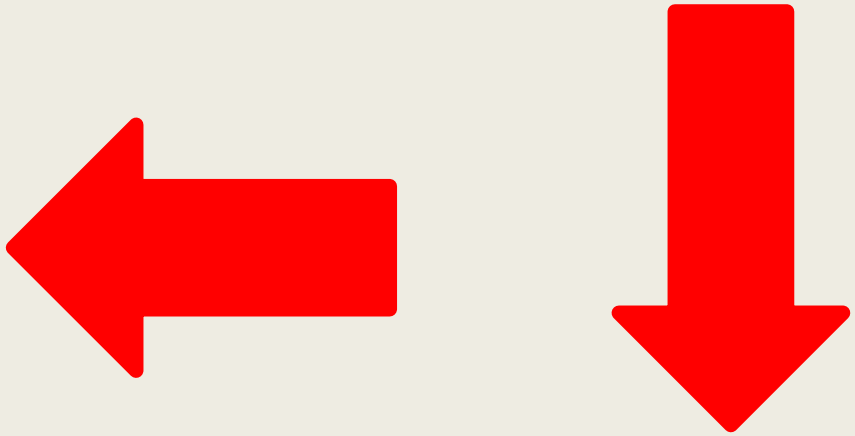
- Timetables



Sample student timetable:

BlackBoard

-



ASSESSMENT

Assessment

- Weekly formative tutorial exercises
  - *Appx. 20 in all*
  - *Unlimited attempts, feedback*
- Two *In-Class tests*
  - *1 hour each **
  - *50% of the module each*
- Further details TBA
- Provisional dates:
  - *Test 1: Week 7 **
  - *Test 2: Week 12 **

* Provisional

Passing the module

- To pass this module, you must:
  - *Achieve at least 40% average*

  AND

  - *Achieve at least 30% in EACH* test

How to pass the module:

- Attend all lectures (Watch recordings after the lecture)
- Do all tutorials as they are released
  - *Keep re-attempting*
  - *Ask for help at tutorials*
- Attend all Seminars
- Do read recommended further reading
- Ask if you struggling
  -
  -
- Help your colleagues if they ask for help
- Use your Personal Tutor
- Sit the tests when they happen

Thank you

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